

Weekly Progress Report

Amity School of Engineering & Technology

Project Group ID - 265

**Project Area – Data Science
(Machine Learning)**

B. Tech (8CSE-DS-1Y) VII Semester

WPR No. 6

Duration: 12/01/2026 to 18/01/2026

Students Weekly Progress Report (WPR) for Even Semester of session 2025-2026

To be filled by Students	
Students Name	1. Sagar Singh Bisht 2. Aakarsh Pab
Roll no.	1. 056 2. 050
Enrollment no.	1. A023167022056 2. A023167022050
Project Title finalized, if Yes, give name, if NO, give reason	Machine Learning based Predictive Model for Livability using Environmental and Socioeconomic indicators
Synopsis submitted	YES
Literature review	YES
Technical & Economical Feasibility	YES
Bill of Material	NA
Project Progress Schedule (PERT Chart)	NA
Design of critical components	-
Fabrication work (give %)	-
Experimental work (give %)	-
Result and Analysis	-
Report writing	-
Signature of students	

Work done in this week:

1. Objective of the Week

To design and implement the **Content-Based Filtering Logic** (The Recommender Engine). The primary goal was to translate the cleaned data and clusters into a functional query system that accepts a user's specific constraints (e.g., "Max Rent: ₹20k", "Priority: Clean Air") and returns a ranked list of neighborhoods using **Vector Space Modeling**.

2. Tasks Completed

A. Recommender Algorithm Implementation

- **Logic Selected:** Cosine Similarity (Vector Space Model).
- **Implementation:**
 - Constructed the Neighborhood Vectors (V_n) from the normalized feature set (Rent, Air, Traffic, Amenities).
 - Constructed the User Query Vector (V_u) dynamic input.
 - Programmed the similarity function:

$$\text{Similarity} = \cos(\theta) = \frac{V_u \cdot V_n}{\|V_u\| \|V_n\|}$$

- **Outcome:** The system can now mathematically rank every neighborhood from 0 to 1 based on how close it is to the user's "Ideal Neighborhood."

B. Feature Inversion Engineering

- **Problem:** Standard similarity algorithms assume "Higher Value = Better." However, for attributes like Rent, Pollution, and Traffic, a lower value is desirable.
- **Action:** Implemented a Directional Transformation Layer before the similarity calculation.
 - *Positive Attributes (Schools, Metro):* Kept as is (Scale 0-1).
 - *Negative Attributes (Rent, PM2.5):* Inverted using the formula $f(x) = 1 - x_{\text{scaled}}$.
- **Result:** Ensures that when a user asks for "High Priority on Rent," the system correctly prioritizes *cheaper* areas.

C. Weighted Priority Logic

- **Action:** Developed the algorithm to handle User Weights.
- **Mechanism:** Instead of treating all features equally, the User Vector is multiplied by a "Priority Matrix."

- *Example:* If a user sets "Air Quality" slider to 10 and "Metro" to 2, the Air Quality feature contributes 5x more to the final similarity score than the Metro feature.

D. Unit Testing (Backend)

- **Action:** Wrote Python scripts to test the logic against known edge cases without a UI.
- **Test Case 1:** "The impossible request" (Lowest Rent + Best Air + South Delhi Location).
 - *Result:* The system successfully returned the "Best Compromise" (e.g., a green area in East Delhi) rather than crashing or returning zero results.

3. Technical Challenges & Solutions

Challenge	Solution
Zero-Vector Error	If a user left all priorities at '0', the Cosine Similarity calculation attempted to divide by zero.
Granularity Loss	The "Inversion Logic" made extremely expensive areas (Rank 0) look too similar to moderately expensive areas (Rank 0.2).

4. Plan for Next Week (Week 4)

Dashboard Development (Frontend Phase 1):

- Initialize the Streamlit web application framework.
- Design the Sidebar Layout for the priority sliders.

Integration:

- Connect the Python Recommender Logic (Backend) to the Streamlit Front-end to test real-time interaction.

To be filled with a Guide (strike off whichever is not applicable)	
The performance of students is satisfactory	
The performance of students is unsatisfactory	
A warning to be issued to student(s) (Name)	
Student was not well (Name)	
Date 12/08/2025	Signature of Guide